Financial metrics

Every metric on the Lightyear dashboard and in the Excel workbook: where it is, the formula, and how to read it.

> **How this is arranged.** The sections follow the dashboard from top to bottom, one section per card, with the card's own name. Under each heading two lines say where to find it: the dashboard card, and the workbook sheet. **Part 1** is the main display, on monthly returns. **Part 2** is the daily calculations and the run diagnostics, both collapsed on the page. **Part 3** is the Gate B tab.
>
> In the workbook, a **formula** means Excel calculates the figure itself; a **pasted value** means Python calculated it and the workbook only shows it. There are no run results or examples here.

Notation used in every formula

Symbol	Meaning
r, b	One day's return on the strategy and on the benchmark, the NIFTY 500 price index.
m	One calendar month's return, compounded from the daily returns. Used throughout Part 1.
Rf (5%)	The risk-free rate in every Sharpe and Sortino. One config setting. A yardstick, not income.
c	The card's cash rate: what idle cash earns in the backtest. Beta and the factor regression subtract it. Not the risk-free rate.
V_t	Portfolio value on day t.
mean, stdev	Average and sample standard deviation (divides by n − 1). **Pop stdev** divides by n; Part 1 uses it.
√252, √12	Annualise daily and monthly volatility. 252 is the assumed trading days a year.
years	Calendar days ÷ 365.25. In Part 1, months ÷ 12.

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Part 1. The main display (monthly)

> **Everything in Part 1 is on monthly returns.** Calendar-month returns m, the population standard deviation, CAGR by month count, and three inputs: risk-free rate 5%, Omega threshold 1% a month, drawdown limit 0%. **Complete calendar months only:** a month counts when the run has a return for every trading session in it; the first month also counts when the only session without a return is its first, the run's starting point. Days outside complete months are left out and listed in the notes at the foot of the page. Python and Excel calculate each figure separately, and the sheet "Python vs Excel" marks each OK or FAIL at 1e-9. The benchmark is the NIFTY 500 price index until a total-return series is on file.

1.1 Strategy vs benchmark: the five tiles

> **Dashboard:** Results tab, card "Strategy vs benchmark (monthly)".
> **Excel:** main workbook, sheet "Returns overview", rows 6 to 10, links to "SE Return Analytics".

The tiles are CAGR and Total Return (formulas in 1.5) and Annualized Volatility, Sharpe Ratio and Max Drawdown (formulas in 1.9), for the strategy with the benchmark under each. The notes at the foot of the page give the first and last complete month and the daily returns left out.

1.2 Financial-year returns

> **Dashboard:** Results tab, card "Financial-year returns".
> **Excel:** main workbook, sheet "Returns overview", second table.

Metric	Formula	How to read it
Financial Year Performance	∏(1 + m) from April to March − 1	* marks the first year, ** a last part-year. A part year is not annualised.
Alpha, each year	Strategy's year − benchmark's year	Points ahead of the index in that year.
Annualised alpha	$CAGR - CAGR_{benchmark}$	The Alpha row of 1.11, shown under the table.

1.3 Strategy against index NAV

> **Dashboard:** Results tab, card "Strategy against index NAV (100 at start)".
> **Excel:** main workbook, chart on "Returns overview"; values on "Monthly series", columns E and F, formulas.

Chart	Calculation	How to read it
NAV, 100 at start	100 × running ∏(1 + m), month-end values	Above the benchmark's line means ahead of it at that month-end.

1.4 Trailing returns

> **Dashboard:** Results tab, card "Trailing returns".
> **Excel:** main workbook, sheet "Returns overview", third table, formulas over "Monthly series".

Metric	Formula	How to read it
1M, 3M, 6M, 1Y	∏(1 + m) over the last 1, 3, 6 or 12 months − 1	Not annualised.
3Y, 5Y, 10Y, since inception	$(\prod(1 + m)$ over the last k months$)^{12 \div k} - 1$	Annualised. A period longer than the run shows n/a.
Alpha	Strategy's figure − benchmark's	

1.5 Key return metrics

> **Dashboard:** Results tab, card "Key return metrics".
> **Excel:** main workbook, sheet "Returns overview", fourth table, links to "SE Return Analytics" column K.

Metric	Formula	How to read it
Total Return	∏(1 + m) − 1	What 1 became over the whole window.
CAGR	$(1 + \text{Total Return})^{12 \div \text{months}} - 1$	Twelve months count as a year, whatever the calendar days.
Avg Month Ret	mean(m)	A typical month.
Positive Months	count of months with m > 0	
Negative Months	count of months with m ≤ 0	A flat month counts as negative.
Positive Alpha Months	count of months with m > b	Months ahead of the index.
Hit Rate	Positive Monthly Alpha ÷ (Positive + Negative Monthly Alpha)	Share of months ahead; ties are left out.
Best Month Ret	max(m)	
Worst Month Ret	min(m)	
Best Month Contribution	Max Month Ret ÷ Total Return	How much of the whole gain one month is worth.
Worst Month Contribution	Min Month Ret ÷ Total Return	

1.6 Market phases

> **Dashboard:** Results tab, card "Market phases": the shaded chart, then the table.
> **Excel:** main workbook, sheet "Market phases", formulas over "Monthly series".

Metric	Formula	How to read it
Phase	Whole calendar months from the phase table in config/market_phases.yaml; the first phase is clipped to the run	A phase with no end month is still running.
Absolute return	∏(1 + m) over the phase's months − 1	The strategy's figure. The chart is the strategy's NAV, and labels each phase with its name, its dates and this figure.
Annualised return	$(1 + \text{absolute return})^{12 \div \text{months}} - 1$	**Annualised whatever the length**, so a two-month phase is scaled up sixfold in the exponent.
Length (months)	Count of the run's months in the phase	

1.7 Rolling returns

> **Dashboard:** Results tab, card "Rolling returns": two charts, then the numbers.
> **Excel:** main workbook, sheet "Rolling returns"; the series are on "Monthly series", columns M to X, formulas.

Metric	Formula	How to read it
Rolling 1-year return	∏(1 + m) over the 12 months ending at each month − 1	
Rolling 3-year return	$(\text{∏(1 + m) over the 36 months ending at each month})^{1 \div 3} - 1$	Annualised.
Rolling alpha	Strategy's rolling return − benchmark's	

Metric	Formula	How to read it
Rolling Sharpe	(rolling return − Rf) ÷ (pop stdev of the window's months × √12)	
Excess Sharpe	Strategy's rolling Sharpe − benchmark's	**An assumed definition**, labelled as such on the page.
Share of observations with positive alpha	Count of windows with alpha > 0 ÷ count of windows	
Average, minimum, maximum rolling return	mean, min and max of the rolling returns, strategy and benchmark	

1.8 Returns attribution to factor indices

> **Dashboard:** Results tab, card "Returns attribution to factor indices": three tables, then the rolling chart.
> **Excel:** main workbook, sheet "Factor attribution", LINEST formulas; the indices' monthly returns are on "Monthly series", columns I to L (pasted), rolling coefficients in Y to AB.

> **No alpha term.** The regression has no intercept, so it only splits the return across the indices. Gate B's factor test (3.2) is a separate regression: daily returns, seven factors, with an intercept.

Metric	Formula	How to read it
Coefficients	Least squares of m on the three indices' monthly returns, through the origin	The indices: NIFTY500 Quality 50, NIFTY500 Momentum 50 and NIFTY Smallcap 250, price indices. Value 50 is left out: its coefficient is not different from zero beside the other three.
R Square; Multiple R	Σ fitted² ÷ Σ m²; its square root	The uncentred R Square a no-intercept regression reports.
Adjusted R Square	1 − (1 − R Square) × n ÷ (n − k)	n is the number of months, k the number of indices (3).
Standard Error	√(Σ residual² ÷ (n − k))	
ANOVA: SS, MS, F, Significance F	Regression SS = Σ fitted², residual SS = Σ residual², MS = SS ÷ df, F = $MS_{regression}$ ÷ $MS_{residual}$, with k and n − k degrees of freedom	
Standard error, t stat, P-value, 95% range	se from the residual variance; t = coefficient ÷ se; two-sided p from Student's t with n − k degrees of freedom; coefficient ± $t_{0.975}$ × se	
One-year rolling attribution	The same coefficients on each 12-month window	Twelve months in each window, so they move a great deal.

1.9 Drawdown and advanced risk ratios

> **Dashboard:** Results tab, card "Drawdown and advanced risk ratios": the chart, then the table.
> **Excel:** main workbook, sheet "Drawdown and risk", links to "SE Return Analytics"; the drawdown series is on "Monthly series", columns G and H.

Metric	Formula	How to read it
Monthly drawdown *(chart)*	W ÷ highest W so far − 1 at each month-end, W = running ∏(1 + m)	0% is a new month-end high.
Annualized Volatility	pop stdev(m) × √12	Lower is smoother. Population stdev, on months.

Metric	Formula	How to read it
Negative Monthly Vol	pop stdev of the months with m ≤ 0	Bumpiness of the losing months. Not annualised.
Downside Deviation	Negative Monthly Vol × √12	The annualised figure. The reference deck prints this one under the name Negative Monthly Vol.
Largest Monthly Loss	Worst Month Ret of 1.5	
Max Drawdown	min over months of (W ÷ highest W so far − 1), where W = running ∏(1 + m)	A negative number. **The peak starts at the first month's own value, so a loss in the first month is not counted.**
Average Annual Max Drawdown	mean of the Max Calendar Year Drawdown rows	A typical year's worst fall. The same first-month rule applies inside each year, and part-years are included.
Sharpe Ratio	(CAGR − Rf) ÷ Annualized Volatility	Return above the risk-free rate per unit of risk.
Treynor Ratio	(CAGR − Rf) ÷ Beta	Return above the risk-free rate per unit of market risk.
Sortino Ratio	(CAGR − Rf) ÷ (Negative Monthly Vol × √12)	**Differs from the textbook:** the divisor is the spread of the losing months, not a downside deviation below a target.
Calmar Ratio	CAGR ÷ |Max Drawdown|	Return per unit of worst fall.
Sterling Ratio	CAGR ÷ (|Average Annual Max Drawdown| − drawdown limit)	**Differs from the textbook:** the usual 10-point adjustment is replaced by the drawdown limit, which is 0.
Omega Ratio	Σ(m − T) over months with m > T ÷ |Σ(m − T) over months with m < T|, T the monthly threshold	Above 1 means gains over the threshold outweigh shortfalls.
Gain to Pain ratio	Σ of months with m ≥ 0 ÷ |Σ of months with m < 0|	Winning months against losing months.
Tail Ratio	|95th percentile of m| ÷ |5th percentile of m|	Above 1 means big wins are larger than big losses.
Positive Monthly Vol ***(workbook only)***	pop stdev of the months with m ≥ 0	Bumpiness of the winning months. Not annualised.

1.10 Tail risk

> **Dashboard:** Results tab, card "Tail risk": the table, then three charts of monthly returns with the VaR and CVaR lines. **Excel:** main workbook, sheet "Tail risk": strategy and benchmark link to "SE Return Analytics"; the alpha column is formulas over "Monthly series" column D.

Three columns: the strategy, the benchmark, and the monthly alpha m − b. Each row is the same formula on that column's months.

Metric	Formula	How to read it
Distribution chart	Count of months in each 2-point band of monthly return, strategy and benchmark	The left end is the tail; lines mark the strategy's VaR and CVaR.
Mean	mean of the months	
Volatility (monthly)	pop stdev of the months	Not annualised.
Skewness	sample skew of m	Negative means big falls are more common than big rises.
Kurtosis	sample excess kurtosis of m	A bell curve scores 0. Positive means fat tails.
95% VaR	5th percentile of m	A negative return, not a positive loss.

Metric	Formula	How to read it
99% VaR	1st percentile of m	A negative return.
95% CVaR	mean of the months with m ≤ 95% VaR	The average month once you are in that tail.
99% CVaR	mean of the months with m ≤ 99% VaR	

1.11 Performance versus index and capture

> **Dashboard:** Results tab, card "Performance versus index and capture": three tables, then the quarterly table.
> **Excel:** main workbook, sheet "Index and capture", links to "SE Return Analytics"; quarterly returns are in its columns F and G.

Performance Vs Index

Metric	Formula	How to read it
Beta	slope of m on the benchmark's monthly returns	1.0 moves in line with the index.
Alpha	CAGR − $CAGR_{benchmark}$	Points a year ahead of the index.
Jensens Alpha (Adj Beta)	CAGR − $CAGR_{benchmark}$ × Beta	**Differs from its label:** no risk-free rate and no adjusted beta are used.
Positive Monthly Alpha	count of months with m > b	Months ahead of the index.
Negative Monthly Alpha	count of months with m < b	Months behind it.
Tracking Error	pop stdev(m − b) × √12	How far it strays from the index.
Information Ratio	Alpha ÷ Tracking Error	The CAGR gap, not the mean monthly gap, over tracking error.
Hit Rate	Positive Monthly Alpha ÷ (Positive + Negative Monthly Alpha)	Share of months ahead; ties are left out.
Max Monthly Alpha	max(m − b)	
Min Monthly Alpha	min(m − b)	

Capture Ratio

Metric	Formula	How to read it
Ret +Ve	mean(m) in months with b ≥ 0	What it earned in the index's up months.
Ret -Ve	mean(m) in months with b < 0	What it lost in the index's down months.
Upside Capture	Ret +Ve ÷ the benchmark's Ret +Ve	Above 1 rises more than the index.
Downside Capture	Ret −Ve ÷ the benchmark's Ret −Ve	Below 1 falls less than the index.
Capture Ratio	Upside Capture ÷ Downside Capture	Above 1 gains more than it gives back.

Extreme Capture Ratio

Metric	Formula	How to read it
Ex. Ret +Ve	mean(m) in months with b ≥ 5%	The index's strongest months.
Ex. Ret -Ve	mean(m) in months with b < −5%	The index's worst months.
Ex. Upside Capture	Ex. Ret +Ve ÷ the benchmark's	
Ex. Downside Capture	Ex. Ret −Ve ÷ the benchmark's	
Extreme Capture Ratio	Ex. Upside Capture ÷ Ex. Downside Capture	Whether it holds up when the market moves hardest.

Average quarterly performance

Metric	Formula	How to read it
Quarterly return	∏(1 + m) over the calendar quarter − 1	Complete quarters only; a part quarter is left out and named.
Average return in rising / falling quarters	mean of the quarterly returns in quarters with the benchmark's return ≥ 0 / < 0	
Upside / downside capture	Strategy's average ÷ benchmark's average, in rising / falling quarters	Above 1 up and below 1 down is the good shape.
Capture ratio	Upside capture ÷ downside capture	

1.12 Calendar-year tables

> **Dashboard:** Results tab, card "Calendar-year performance and max calendar-year drawdown".
> **Excel:** main workbook, sheet "Calendar years", links to "SE Return Analytics".

Metric	Formula	How to read it
Calendar Year Performance	∏(1 + m) from January to December − 1	* marks the first year, ** a last part-year.
Max Calendar Year Drawdown	The Max Drawdown formula on that year's months alone	**A loss in the year's first month is not counted**, because the peak starts at that month's own value.

1.13 Market-cap and sector allocation

> **Dashboard:** Results tab, card "Market-cap and sector allocation and returns, by financial year".
> **Excel:** main workbook, sheet "Allocation": yearly tables are formulas over the per-rebalance figures to their right, which are pasted.

Metric	Formula	How to read it
Bucket	Large = Nifty 100, mid = Midcap 150, small = Smallcap 250, from the latest quarterly NIFTY 500 list on or before the rebalance. Sector = Ace's present-day sector, grouped by config/sector_groups.yaml	Sectors are not dated: each company is under its present sector for every year.
Allocation	Average, over the year's rebalances, of the bucket's weight	Cash is 1 − the sum of the buckets.
Market-cap return	Holdings' returns from one execution date to the next, weighted by their weights, compounded over the year	A bucket holding one small position takes that stock's return.
Sector return	Plain average of the sector's holdings' returns per holding period, compounded over the year	

Part 2. Daily calculations and run diagnostics

Both sections are collapsed on the Results tab, under the main display. The daily figures use every daily return, the sample standard deviation and 252 days a year, so they differ from the monthly figures of the same name in Part 1. Gate B reads the run diagnostics and its own daily tests.

2.1 Daily calculations

> **Dashboard:** Results tab, section "Daily calculations": one row per metric, with its daily formula, the daily value and the monthly figure beside it.
> **Excel:** main workbook, sheet "Daily calculations", formulas over the "Returns" sheet, checked on "Python vs Excel". The "Summary" sheet of the diagnostics workbook repeats the first rows.

Metric	Formula	How to read it
CAGR	$(V_{end} \div V_{start})^{1 \div years} - 1$, with $V_{start} = 100$ on the session before the first return	The steady yearly growth rate from start to finish.
Volatility	stdev(r) × √252	How much returns swing in a year. Lower is a smoother ride.
Sharpe (risk-free rate 5%)	(CAGR − 5%) ÷ Volatility	Return above the risk-free rate, per unit of risk.
Maximum drawdown	max over t of $(1 - V_t \div$ highest V so far$)$	The biggest loss from a high to a later low, as a positive number.
Final value	100 × ∏(1 + r)	What 100 grew into after every daily return.
Sortino (risk-free rate 5%)	(CAGR − 5%) ÷ DD, where DD = √mean(min(r − Rf per day, 0)²) × √252	Like the Sharpe, but only falls below the risk-free rate count as risk.
Calmar	CAGR ÷ maximum drawdown	Return per unit of worst fall.
Drawdown duration	Longest run of trading days below a previous high	How long the longest wait to recover took.
95% CVaR (daily loss)	−mean of the daily returns at or below their 5th percentile	The average bad day, as a positive number.
Skew	Sample skew of daily returns	Negative means big falls are more common than big rises.
CAGR without the 10 best days	Remove the 10 best days, then compound; years run from the first return date	Whether a handful of days made the result.
Beta	Cov(r − c, b − c) ÷ Var(b − c)	1.0 moves in line with the index. Below 1 is less sensitive.
Tracking error	stdev(r − b) × √252	How far it strays from the index.
Information ratio	mean(r − b) × 252 ÷ tracking error	Return over the index per unit of straying. Higher is better.
Rolling one-year periods behind	Share of 252-day windows with compounded return below the index's	How often an investor would have been trailing.
Two-way turnover a year *(Run facts card)*	Σ |weight after − weight before| over every trade, ÷ years	1.0 means trading the whole portfolio's value once a year.
After publication *(Run facts card)*	CAGR, benchmark CAGR, their gap (edge), arithmetic Sharpe and maximum drawdown, on the days from the paper's publication date	Whether the result held once the idea was public. A date given only as a year is read as 31 December.
Capacity per Rs 1 crore *(Run facts card; dashboard only)*	For each trade: (its share of capital × Rs 1 crore) ÷ the stock's 20-day average traded value. The 95th percentile P is shown, and the capital at which it reaches the limit: 10% ÷ P	Information only. No fund size is assumed, and the limit is a placeholder.

Metric	Formula	How to read it
Sharpe by calendar year, arithmetic *(Run facts card; dashboard only)*	mean(r − Rf per day) ÷ stdev(r − Rf per day) × √252, on that year's days	The yearly form of the Sharpe. It is labelled arithmetic wherever it appears.

Beta in different markets (card "Run facts (daily)"; dashboard only)

Metric	Formula	How to read it
Up-market / down-market beta	Beta using only days the index rose / fell	Ideally high in rises and low in falls. The two do not average to the overall beta.
Index filter open / shut	Beta using only days the card's index filter was open / shut. The filter's state on a day is read at the previous session's close	Shown when the card has an index filter. A low beta when shut is the filter holding cash.

2.2 The daily charts

> **Dashboard:** Results tab, section "Run diagnostics", one card per chart (daily).
> **Excel:** diagnostics workbook, sheet "Returns": value in columns D and E, drawdown in F and G, formulas. Sheets "Calendar-year returns" and "Monthly returns".

Chart	Calculation	How to read it
Growth of 100	100 × running product of (1 + r)	Above the benchmark's line means ahead of it on that day.
Drawdown	V_t ÷ highest V so far − 1	0% is a new high. Deeper dips are bigger losses.
Rolling 3-month volatility	stdev of the last 63 daily returns × √252	How bumpy the last three months were. Dashboard only.
Calendar-year return	Product of (1 + r) within the year − 1	First and last years may be part-years.
Strategy monthly returns	Product of (1 + r) within the month − 1	Green months gained, red months lost.

2.3 Appendix: the simpler alternatives

> **Dashboard:** Results tab, section "Run diagnostics", card "Appendix: the simpler alternatives it had to beat".
> **Excel:** diagnostics workbook, sheet "Comparators": daily returns from column H, the metrics table is formulas. Sheet "Performance" holds the same figures as pasted values.

For each alternative (for example the equal-weight basket) the card shows CAGR, volatility, Sharpe and worst fall. They are the formulas in 2.1 on that alternative's daily returns over the same days.

Sections 2.4 to 2.8 are the five cards of rows written by each run's own script. **Rows tied to a strategy's rules appear only for that strategy**, so another run can show a different list.

2.4 Returns & risk

> **Dashboard:** Results tab, section "Run diagnostics", card "Returns & risk".
> **Excel:** diagnostics workbook, sheet "Validation", pasted values (blue), in the same order.

Metric	Formula	How to read it
CAGR, volatility, Sharpe, max drawdown: strategy, index, alternatives	The formulas in 2.1	The same headline figures, listed for each series.
Edge over the index	$CAGR_{strategy} - CAGR_{benchmark}$	Points a year ahead of the index.
Drawdown in the latest index correction	Each series' fall over the index's latest fall deeper than 10%	How it behaved in the most recent real test.

Metric	Formula	How to read it
Beta; information ratio	As in 2.1	
Calendar-year return, a named year	Product of (1 + r) within that year − 1	A year the run's notes single out.
Deflated Sharpe, at V = 1 and at V from the variants	$P(\text{skill}) = \Phi[(SR - SR_0) \times \sqrt{(n-1)} \div \sqrt{(1 - S \cdot SR + (K-1) \div 4 \times SR^2)}]$, with the luck bar $SR_0 = \sqrt{V} \times [(1-\gamma)\,\Phi^{-1}(1 - 1 \div N) + \gamma\,\Phi^{-1}(1 - 1 \div (N\,e))]$. SR is the arithmetic Sharpe per day, n the days, S and K skew and kurtosis, N the versions tried, V the variance of their Sharpes	The chance the Sharpe is skill, after allowing for N versions tried. Very sensitive to V, so two answers are shown. It is outside Gate B.

2.5 Costs & execution

> **Dashboard:** Results tab, section "Run diagnostics", card "Costs & execution".
> **Excel:** diagnostics workbook, sheet "Validation", pasted values (blue), in the same order. The sweep itself is on the sheet "Appendix cost sweep".

Metric	Formula	How to read it
Two-way turnover a year	As in 2.1	
Breakeven round-trip cost	The cost at which CAGR equals the benchmark's, read between two points of the cost sweep	How much trading cost the edge can absorb. Blank if beyond the sweep.
Edge at a high cost	CAGR re-simulated at that round-trip cost − $CAGR_{benchmark}$	What survives expensive trading.
Fills over the run	Count of trades: buys, sells and resizes	How busy the book is.
Names held per rebalance, mean	mean(stocks held per rebalance)	Too few means the result rests on few companies.
Largest single weight	max(single weight at any rebalance)	Concentration risk.
Different stocks held	Count of distinct names	Higher means more breadth.
Top-2 stocks' share of gains	Gains of the two best names ÷ gains of all winners	High means a couple of stocks made the result.
Mean / maximum cash share	mean / max(cash ÷ NAV)	How invested the book really was.
Median holding period	Median calendar days, entry to exit	How long a position typically lasts.
Small-cap share	mean(NAV share in Smallcap 250 names)	How much of the result is a small-cap bet.

2.6 Statistical robustness

> **Dashboard:** Results tab, section "Run diagnostics", card "Statistical robustness".
> **Excel:** diagnostics workbook, sheet "Validation", pasted values (blue), in the same order. Sheets "Sensitivities", "Cash-rate sweep", "Sub-period stability" and "Sub-period CAGR" hold the tables behind them.

Every Sharpe in this card is arithmetic: mean(r − Rf per day) ÷ stdev(r − Rf per day) × √252.

Metric	Formula	How to read it
Sharpe (arithmetic)	mean(r − Rf per day) ÷ stdev(r − Rf per day) × √252	The form the statistical tests use.
Bootstrap Sharpe 95% low / high	2.5th and 97.5th percentile of the Sharpe across 2,000 re-shuffled histories (stationary bootstrap, blocks of days)	A range that includes low values means the Sharpe is uncertain.

Metric	Formula	How to read it
Paired P(no Sharpe edge), each comparator	Share of re-shuffled histories where strategy Sharpe − comparator Sharpe ≤ 0, both cut at the same days	Small means the edge is hard to explain by chance.
Paired P with a 5-day / 63-day block	The same P with the average block length fixed	A check that the answer does not hang on the block length.
Paired Sharpe difference, 95% interval low	2.5th percentile of the Sharpe difference across the histories	Below 0 means no edge is within the range.
Factor alpha a year	Regress r − c on the factor indices' returns − c; α is the intercept. Shown as $(1 + \alpha)^{252} - 1$	Return left after known factors.
Alpha HAC t-stat; p-value	α ÷ its HAC (Newey-West) standard error; p is the two-sided probability of that t	How sure we are alpha is not zero.
Factor alpha over the risk-free rate	The same regression with returns above Rf instead of c	Shown beside the main figure.
Loading on a factor	β_k in the factor regression	How much of that factor it behaves like.
Factor R-squared	Share of variance the factors explain	High means mostly known factors.
Alpha with a factor left out; on a variant	The same regression with one factor removed, or on a variant book	Shows how much one factor or rule was accounting for.
Worst sub-period excess CAGR	The smallest $CAGR_{strategy} - CAGR_{benchmark}$ across the windows	Whether the result depends on one stretch.
Last window: CAGR, Sharpe	CAGR and arithmetic Sharpe on the last window's days	How it did most recently.
CAGR after one rule change	Re-simulate with one rule changed	A real effect keeps its sign when a rule moves.
CAGR / edge from a later start	Re-simulate from that start date	Tests the years with the weakest data.
CAGR with cash at another rate	Re-simulate with idle cash earning that rate	How much the cash rate contributes.
Deflated Sharpe: selection threshold	The luck bar SR_0 in 2.4, as a yearly Sharpe	The Sharpe luck alone could reach after N versions.

2.7 Diagnostics

> **Dashboard:** Results tab, section "Run diagnostics", card "Diagnostics (specific to this strategy's rules)".
> **Excel:** diagnostics workbook, sheet "Validation", pasted values (blue), in the same order. Sheets "Rebalance log", "Exits by reason" and "Exits (every sell)" hold the detail.

Metric	Formula	How to read it
Rebalances with few names	Count with fewer than 5 names, overall and while buying was open	Separates a filter holding cash from a universe running dry.
Signals with buying shut; share	Count of signals with the index filter closed; ÷ all signals	How often the market filter stopped new buys.
Signals fully in cash; with few eligible names	Counts of signals	How often filters left nothing or little to buy.
Eligible names, mean per signal	mean(names passing every filter)	How wide the choice was.
Names removed by a rule	mean(top-band names rejected per signal)	How hard a filter bites.
Names bought and cash share at the first fill	Count of positions opened on day one; cash ÷ NAV after it	How the book starts.
Resize fills	Count of trades that only resized a position	How much trading is upkeep.

Metric	Formula	How to read it
Rebalances with a weight above 25%	Count	Concentration risk.
Exits by rule; their mean return	Count of sells forced by each rule; mean return since entry	Whether a rule mostly cuts losers or winners.
Sells on leaving the index; their mean return	Count; mean return since entry	What index exits cost or earned.
Buys / sells on big-move days	Count of trades after a close up or down more than 9.5%	Proxy for trading at a price band, where a real fill may fail.
Suspensions and delistings sat through	Count of held names with no price	How often the missing-price rule was used.

2.8 Other

> **Dashboard:** Results tab, section "Run diagnostics", card "Other".
> **Excel:** diagnostics workbook, sheet "Validation", pasted values (blue), in the same order. Sheets "Checks" and "Universe coverage" hold the detail.

Row	What it counts
Checks passed	Look-ahead and accounting checks that passed, out of those run.
Signals; trading days; last signal and fill; run end date	How many decisions and days the run covers, and where it stops.
Universe source; members; members priced; never priced	Which index list was used, and how many members could actually be traded.
Price moves that coincide with an adjustment change	Whether a price rule is tripped by splits and bonuses rather than real shocks.
Trials for the deflated Sharpe	N in 2.4: versions from the paper, earlier runs and studies, plus this run's sensitivities.
Previous run rows	The same metrics copied from the earlier run, for comparison.

Part 3. The Gate B tab

> **Dashboard:** Gate B tab, card "Gate B: investment decision" (gate version v4): the tables "Hard blocks" and "Evidence scorecard".
> **Excel:** diagnostics workbook, sheet "Gate B", pasted results. The status is repeated on its "Summary" sheet.

Hard blocks decide the status, PASS or FAIL. Scorecard rows are green, amber or red and do not block. A test with nothing to run on shows as not run, which is not a pass. Every Sharpe here is arithmetic, over the risk-free rate of 5%.

3.1 Hard blocks

Test	What is calculated	Pass mark	Kind
Data integrity	Every check the run wrote passed; no card rule broken; coverage = Σ(priced members × sessions) ÷ Σ(all members × sessions)	All pass; coverage ≥ 90%	Hard block
Edge over the benchmark	$CAGR_{strategy} - CAGR_{benchmark}$, at the cost on the card	> 0	Hard block
Paired bootstrap	P = share of 2,000 re-shuffled histories where $Sharpe_{strategy} - Sharpe_{benchmark} \le 0$. Block length chosen from the data; seed 7	P ≤ 0.1	Hard block

3.2 Evidence scorecard

Test	What is calculated	Pass mark	Kind
Sub-period stability	$Sharpe_{strategy} - Sharpe_{benchmark}$ in each of 4 equal windows; the value shown is the worst gap	Green: ≥ 50% of windows ahead and none below -1.0. Amber: worst ≥ -1.5	Scorecard
Factor alpha, one-sided	Regress r − c on 7 factor indices' returns − c. t = α ÷ its HAC standard error, with lags = $\lceil 4\ (n \div 100)^{2/9} \rceil$	Green: α > 0 and t ≥ 2.0. Amber: t ≥ 1.0	Scorecard
Performance after publication	$CAGR_{strategy} - CAGR_{benchmark}$ from the paper's publication date; needs 252 trading days	Green: > 0%. Amber: > -2%	Scorecard

Under each test the tab says in plain words what it asks. The decision stays PENDING until a named person records it.

The two workbooks

Both are downloaded from the Results tab, and every section above names its sheet. **Main workbook:** Returns overview, Market phases, Rolling returns, Factor attribution, Drawdown and risk, Tail risk, Index and capture, Calendar years, Allocation, Daily calculations, Python vs Excel, SE Return Analytics, Monthly series, Returns. **Diagnostics workbook:** Summary, Returns, Monthly returns, Gate B, Comparators, Validation, Holdings, Rebalances, Trades, and one sheet for each table the run wrote.

Sources: ros/metrics/se.py, ros/metrics/qi.py, ros/metrics/core.py, lightyear/charts.py, lightyear/main_workbook.py, lightyear/allocation.py, lightyear/workbook.py, lightyear/se_sheet.py, lightyear/gate_run.py, ros/governance/gate_v2.py, ros/validation/gate_evidence.py, research.py, portfolio.py, trials.py, config/fund.yaml, config/market_phases.yaml, config/sector_groups.yaml, config/gate_thresholds.yaml.